

Monday Warm Up: Unit 4, Reactance, RL circuits, RLC circuits

Prof. Jordan C. Hanson

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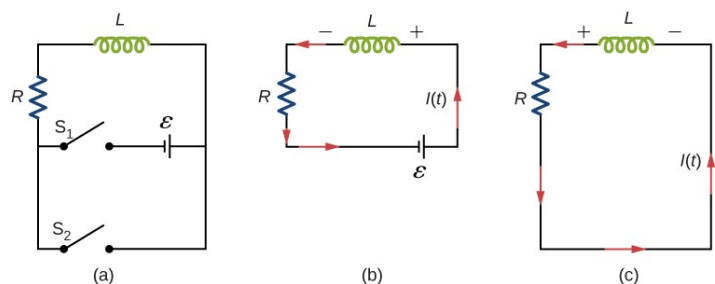


Figure 1: A circuit diagram of a DC emf with a resistor, inductor, and two switches.

2. (a) What is τ when $R = 1 \text{ k}\Omega$, and $L = 0.2 \text{ Henrys}$?
(b) What will the time constant τ be if we double R and quadruple L ?

3. If switch 1 is open and switch 2 closed after $t \gg \tau$, what time passes before the current is reduced to half of its original value?

1 Memory Bank

- $\epsilon = -L \frac{\Delta I}{\Delta t}$... Faraday's law with induction, L . Think of this as the change in voltage across an inductor.
- $X_C = 1/(2\pi fC)$... The reactance of a capacitor with capacitance C , at frequency f .
- $X_L = 2\pi fL$... The reactance of an inductor with inductance L , at frequency f .
- $Z_{\text{tot}} = \sqrt{R^2 + (X_L - X_C)^2}$... The total impedance of an RLC series circuit.
- $\tau = RC$... The time constant of an RC circuit.
- $f_0 = \frac{1}{2\pi\sqrt{LC}}$... RLC circuit resonance frequency.
- $i(t) = \frac{\epsilon}{R} (1 - \exp(-t/\tau))$... Current in a charging RL circuit.
- $i(t) = \frac{\epsilon}{R} \exp(-t/\tau)$... Current in a discharging RL circuit.
- $\tau = L/R$... Time constant of an RL circuit.
- $V_{\text{rms}} = I_{\text{rms}} Z_{\text{tot}}$... Ohm's Law for AC circuits.

2 RL Circuits

1. Consider Fig. 1. There is a battery connected to a resistor R and an inductor L via switch 1. Switch 2 connects L and R . (a) What is the current at $t = 0$, when switch 1 is closed? (b) What is the current when $t \gg \tau$? (This is like part (b) of Fig. 1).

3 Reactance and Impedance

1. (a) What is the reactance, in ohms, of a $2.5 \mu\text{F}$ capacitor at 1400 kHz ? (b) What is the reactance of a 5.0 mH inductor at this same frequency? (c) Now add (in series) a $1 \text{ k}\Omega$ resistor. What is the total impedance at this frequency? (d) What is the total impedance at 1.4 MHz ?
2. What is the resonance frequency of the RLC circuit in the previous exercise?